IPA as a Unified Representation for Multilingual ASR

Ignazio Sorrentino*, Marco Matassoni, Roberto Gretter, Daniele Falavigna and Alessio Brutti

Fondazione Bruno Kessler, Augmented Intelligence Center, via Sommarive 18, Trento, 38123, Italy

Abstract

In this work, we investigate IPA-based multilingual automatic speech recognition using a fine-tuned XLSR-53 phonetic model trained on a curated subset of IPAPACK++. The resulting model overcomes some limitations of traditional grapheme-based systems in case of languages with limited linguistic resources. Our phonetic models, trained on 21 European languages, achieve results comparable to models trained on substantially larger datasets. We evaluate these models in supervised, zero-shot and out-of-domain settings, including test sets from DoReCo, Kazakh, and Tatar. In addition, we propose a hybrid speech-to-IPA-to-grapheme pipeline combining the phonetic model with a Weighted Finite-State Transducer decoder built from text-only resources, achieving a WER of 17.7% on English and 30.3% on a combined multilingual graph covering five European languages. The investigated architecture is particularly suitable for low-resource scenarios, where written corpora are usually much more abundant than transcribed audio.

Keywords

phoneme recognition, multilingual ASR, zero-shot transfer learning, WFST

1. Introduction

Automatic Speech Recognition (ASR) is the task of converting spoken language into written text. The rise of End-To-End (E2E) models has made it possible to achieve human level performance with audio transcriptions [1]. Large amounts of transcribed speech are needed to achieve such results, but the vast majority of languages lack this kind of resource; if less than 100 hours of labeled audio is available, they are usually described as low-resource languages [2, 3, 4].

Popular strategies for addressing data scarcity are self-supervised [5, 6, 7] and semi-supervised learning [8, 9], which require less labeled data than supervised approaches, and multilingual modeling [10, 11], which, on the other hand, can take advantage of the transfer learning capability between languages. Zero-shot transfer learning has also been explored [12, 13]: unseen languages are transcribed with a single multilingual model trained on several languages.

Some research has investigated the use of the International Phonetic Alphabet (IPA) symbols as basic modeling units for ASR rather than the more widely used graphemes. For example, [14] trained an ASR model to recognize IPA token sequences and observed significant improvements across all languages in the multilingual setting. Similarly, [3] demonstrated that an IPA-based phonetic model improves low-resource ASR performance within an attention encoder-decoder framework, outperforming baseline monolingual models.

Building on this line of research, we focus on phonetic-based multilingual ASR, investigating IPA-based universal representations as modeling units. We develop a wav2vec-based phonetic model, which is obtained by fine-tuning XLSR-53 using IPAPACK++ [15], a curated phonetic dataset, to produce IPA transcriptions. For training, we employ 21 European languages[1].

In addition to phoneme recognition, we explore a hybrid approach that combines our phonetic acoustic model with a Weighted Finite State Transducer (WFST) decoder, built with a language model

CLiC-it 2026: Twelfth Italian Conference on Computational Linguistics, September 14–16, 2026, Palermo, Italy

*Corresponding author.

isorrentino@fbk.eu (I. Sorrentino); matasso@fbk.eu (M. Matassoni); gretter@fbk.eu (R. Gretter); falavi@fbk.eu (D. Falavigna); brutti@fbk.eu (A. Brutti)

© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

[1]The languages are: Bulgarian, Czech, Danish, German, Greek, English, Spanish, Estonian, Finnish, Irish, Croatian, Hungarian, Italian, Lithuanian, Maltese, Dutch, Polish, Romanian, Slovak, Slovenian, Swedish.

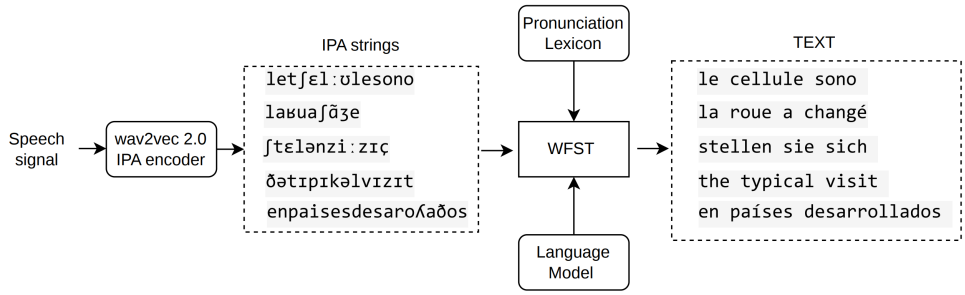


Figure 1: Proposed 2-step pipeline.

(LM) and a pronunciation lexicon (PM). To test the full pipeline, we fine-tune a dedicated IPA model on 5 European languages (English, Italian, German, Spanish, and French) from FLEURS [16], and build language-specific trigram LMs from CommonVoice [17] text data. The five language-specific decoding graphs are then statically linked into a single unified network, enabling the decoder to transcribe all five languages in a single pass without explicit language identification. The phonetic outputs from the neural acoustic model are thus converted into orthographic text via the WFST, completing the speech-to-IPA-to-grapheme pipeline.

Our experiments show that **IPA-based multilingual ASR constitutes a viable and data-efficient alternative** to conventional grapheme-based approaches; we demonstrate competitive performance in supervised, zero-shot, and out-of-domain settings, despite relying on substantially less training data than comparable systems. Furthermore, the proposed **speech-to-IPA-to-grapheme framework** enables multilingual decoding, making it particularly attractive for extending ASR technologies to low-resource languages.

2. State of the Art and Related Work

The ultimate objective of multilingual ASR is to transcribe all spoken languages with a single universal model. Most frameworks, such as Whisper [18] and POWSM [19], aim at transcribing speech directly into graphemes, without considering the unique orthography rules of each language. These models achieve impressive results on high-resource languages [4], but their performance degrades significantly on low-resource and endangered languages, where labeled training data is scarce or unavailable [20].

IPA transcriptions have often been chosen as targets, given their language-agnostic nature. The results in this field have improved with the diffusion of E2E architectures. XLS-R models can be fine tuned to predict IPA targets [12, 13]. [12] fine-tuned XLSR-53 jointly on multiple languages and assessed PR zero shot capacity on unseen languages. Similarly, [13] fine-tuned XLSR-53 on a small but carefully curated set of seven languages, demonstrating that label quality in training outweighs quantity.

More recent work has moved beyond single-task XLS-R fine-tuning to dedicated architectures trained at scale. [15] introduces ZIPA, a family of phone recognition models built on the efficient Zipformer [21] backbone trained on their curated dataset IPAPACK++. This corpus contains 17,132 hours of phonetically transcribed audio and, although the transcriptions were not manually checked by phoneticians, the corpus is more reliable than typical G2P-generated data thanks to a series of systematic curation steps: phone inventories were validated against PHOIBLE [22], inconsistent Unicode encodings were reconciled, and overly complex transcriptions were simplified to at most one diacritic. A complementary direction replaces the encoder-only CTC paradigm with an attention-based encoder-decoder (AED) architecture.

Universal language phoneme representations can be used as a middle layer for two-step ASR models, which first recognize phonemes from speech and then convert them to graphemes [23]. In the field of two-pass ASR with Phoneme-To-Grapheme (P2G) translation, [24] utilizes phonemes as an intermediate representation and applies the two-pass ASR approach for cross-lingual transfer learning. Their study

demonstrates the effectiveness of incorporating phoneme recognition as the initial pass, followed by the translation of recognized phonemes into words (P2W). This approach shows better handling of language-specific variations and improves the performance of the ASR system in general.

3. Method

3.1. Data

Datasets containing audio with transcriptions into IPA are extremely scarce. To overcome this issue, most speech-to-IPA models take advantage of automatic G2P conversion tools to map graphemes to the corresponding sequence of phonemes [12, 25, 26].

For this work, we use FLEURS [16], which provides parallel speech data in 102 languages with approximately 12 hours of speech supervision per language, and CommonVoice [17], a large-scale community-driven corpus of read speech. In order to train a model for IPA symbols, we leverage the FLEURS and CommonVoice sections of IPAPACK++ dataset [15]: this was created by merging G2P-generated transcriptions in 88 languages from various open-source datasets, e.g., FLEURS, CommonVoice, LibriSpeech [27], MultilingualLibriSpeech [28]. Two G2P models were used to automatically retrieve phonemic transcriptions: CharsiuG2P [29], which uses a Transformer-based architecture, and Epitran [30], which is WFST based. Then, PHOIBLE [22] was used to determine whether the IPA symbols were legitimate. Illegitimate phones were corrected and those with more than 3 diactrics were simplified to no more than one. Finally, Unicode normalization was applied. In this way, the noise and errors introduced by automatic G2P generation were reduced, producing more consistent and linguistically valid phonemic transcriptions. This careful normalization is motivated by evidence that labels quality may matter more than the size of the dataset [13, 15].

3.2. Phonetic Model

For training the phone recognizer, we consider the 21 European languages, among the 24 official languages of the European Union, which are present in the FLEURS subset of IPAPACK++. Each language has approximately 10 hours of training data, 1 hour of validation data, and 1 hour of test data. We also use the CommonVoice subset of IPAPACK++ with the same languages. We prepare two different sets of training with different sizes. The first set uses FLEURS exclusively. The second set contains 50 hours of training per language: using FLEURS as a base, CommonVoice samples were added to reach the established threshold.

We first evaluate the IPA models in supervised settings using the FLEURS test sets for the 21 languages seen during the training. In order to also check the universality of the models, we test on 8 unseen European languages from FLEURS and, lastly, we run inference on three out-of-domain datasets: DoReCo [31] (a corpus of small and endengered languages), the Kazakh Speech Dataset [32] and the Tatar CommonVoice test set.

3.3. WFST Decoding

The phonetic output of the IPA model is well suited as middle layer of a two-step pipeline, in which the second element is phoneme-to-grapheme (P2G) decoding. We adopt the *Compact CTC* topology [33] and statically compose the CTC transducer with a pronunciation lexicon FST and a n-gram language model FST. The acoustic model is fine-tuned with a CTC loss over IPA phoneme targets. In inference, the phonetic model produces logarithmic posterior distributions at frame-level over the IPA phoneme vocabulary. For grapheme-level decoding, the same log-posteriors are passed as emission scores to the WFST decoder, where the search is jointly guided by the composed pronunciation lexicon and language model graph, yielding word-level transcriptions.

Language-specific n-gram language models are trained on the CommonVoice text data for each language, compiled into individual FSTs, and linked into a single unified network; as a result, a single

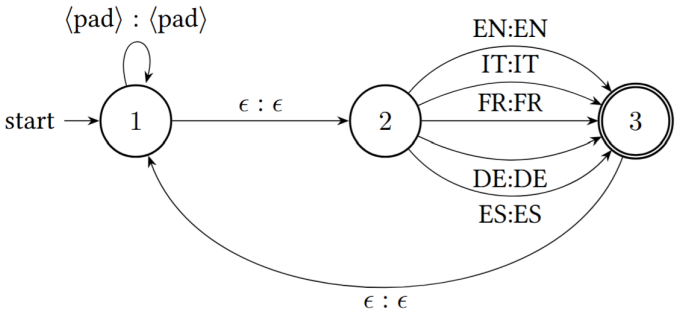


Figure 2: Multilingual WFST. The $\epsilon : \epsilon$ arcs are empty transitions connecting the language-specific subgraphs, while each $L : L$ arc (e.g., IT : IT, EN : EN) represents the composed pronunciation lexicon and n-gram language model FST for language L. The $\langle \text{pad} \rangle : \langle \text{pad} \rangle$ self-loop on the initial state handles CTC padding frames.

decoder pass can handle multiple languages simultaneously, for example, by properly modeling code-switching phenomena. Since the IPA model produces a language-agnostic phoneme stream, the same acoustic front-end can feed multiple language-specific P2G back-ends within one unified graph, making the approach naturally multilingual (see Figure 2).

To test the decoding strategy based on WFST a specific IPA model has been trained on 5 major European languages (English, Italian, German, Spanish, and French), using the same procedure described in Section 3.2.

We then build the WFST decoder for these five languages. In order for the framework to be fully reproducible and consistent, the training set for the IPA model was derived from the phonemization of FLEURS via the eSpeak phonemizer [34]; the same phonemizer is used to derive the prounciation lexica. The individual networks (one per language) are finally linked into a single unified graph capable of handling all languages in a single decoding pass.

4. Experiments

4.1. Setup

We leverage the pretrained model wav2vec2-large-xlsr-53[2], which was trained in a self-supervised manner (SSL) on 56k hours of speech data with 53 languages from MultiLingual Librispeech, CommonVoice and BABEL [35]. We implement the fine-tuned model with Wav2Vec2ForCTC provided in the transformers library to train it with the Connectionist Temporal Classification (CTC) loss [36]. To fine-tune the XLSR-53 model to predict IPA characters, we add a classifier representing the joint vocabulary of the training languages on top of the model and train on the labeled data with a Connectionist Temporal Classification (CTC) loss. The feature encoder is kept frozen during fine-tuning. Training is performed with a batch size of 4 per device and 4 gradient accumulation steps, producing an effective batch size of 16. We use the AdamW optimizer with a learning rate of 3×10^{-5} and 500 warmup steps. The best checkpoint is selected by the PER validation.

For the WFST decoder, trigram language models ($n = 3$) are estimated from the CommonVoice train splits for each of the five languages, then compiled into binary FSTs. Pronunciation lexica are built using eSpeak, mapping each word in the language model vocabulary to its IPA phoneme sequence. The acoustic HMM models follow the Compact CTC topology, with one two-state model per IPA token. The five language-specific FSTs are statically linked into a single unified decoding graph. At inference, frame-level log-posteriors from the IPA model are passed directly to the decoder; no additional language identification step is required, as the language model component implicitly guides the search toward the correct language. We evaluate FLEURS test sets for all five languages.

[2]https://huggingface.co/facebook/wav2vec2-large-xlsr-53

4.2. Evaluation Metrics

The most widely used measures in ASR are Word Error Rate (WER) and Character Error Rate (CER). For Phoneme Recognition the Phoneme Error Rate (PER) is usually adapted; PER is a modification of CER whose basic unit is a phoneme (which may consist of more than one Unicode character) instead of a character.

However, a problem with using PER in this task is that PER ignores phonetic similarities among phones. Consider a model that predicts [pɔnd] for a spoken word labeled [bɔnd], differing only in the first consonant. The phones [p] and [b] are both bilabial stops and differ in a single feature, voicing, so their distance should intuitively be much smaller than that of two completely unrelated phones such as [p] and [ɔ]. This is why PFER (Phoneme Feature Error Rate) has emerged as metric: it weights each substitution according to the phonological distance between the hypothesized and reference phoneme. Rather than treating all substitutions equally costly, PFER leverages phonological feature representations derived from PanPhon [37], a library that maps IPA symbols to articulatory feature vectors. Given n features per vector, PFER calculates the Hamming distance between two phonemes, so that a single feature mismatch has a cost of $1/n$.

5. Results

5.1. IPA In-domain Evaluation

Table 1 compares the PER and PFER scores obtained from fine-tuning XLSR-53 on 21 European languages from the IPAPACK++ dataset. We also evaluate results in a zero-shot setting[3]. The model trained on 50 hours per language (eu21-50h) performs the best in the supervised setting (21.9% PER; 7.5% PFER). In the zero-shot setting, the model trained only on the FLEURS set of IPAPACK++ (eu21-fl) outperforms eu21-50h in PER and achieves a similar result in PFER.

Table 1
PER and PFER (↓%) comparison between our models: eu21-fl refers to the model trained only on 21 languages from FLEURS; eu21-50h is the model trained on 50 hours per language

	PER		PFER	
Model	Supervised	Zero-shot	Supervised	Zero-shot
eu21-fl	24.5	43.3	8.2	12.1
eu21-50h	21.9	44.9	7.5	12.0

In tables 2 and 3 zero-shot results for each language are shown. We achieve the best results with Macedonian (mk) and Galician (gl), with eu21-fl reaching 21.4% and 22.5% PER respectively, while we observe the worst results with Welsh (cy) and Icelandic (is), with PER above 60% for both models.

Table 2
PER (↓%) comparison for zero-shot languages

Model	be	bs	ca	cy	gl	is	mk	nb	ru	sr	uk
eu21-fl	52.9	39.0	38.4	67.1	22.5	66.4	21.4	60.5	59.3	38.3	53.8
eu21-50h	53.0	38.9	38.8	64.9	28.7	68.8	29.5	58.9	58.5	37.1	55.1

[3]The zero-shot evaluation is carried out on: Belarusian, Bosnian, Catalan, Welsh, Galician, Icelandic, Macedonian, Norwegian Bokmål, Russian, Serbian, and Ukrainian.

Table 3
PFER (↓%) comparison for zero-shot languages

Model	be	bs	ca	cy	gl	is	mk	nb	ru	sr	uk
eu21-fl	17.1	9.9	8.3	27.4	6.4	21.6	5.6	18.4	13.1	10.8	11.0
eu21-50h	17.2	10.0	8.4	25.7	7.2	23.9	6.4	15.2	13.3	9.4	11.1

5.2. IPA Out-of-Domain Evaluation

Table 4 compares the performance of our models with other published works, namely Facebook's `wav2vec2-xlsr-53-espeak-cv-ft` [12] and MultiIPA [13]. These are chosen since they were trained with an identical setup, i.e., fine-tuning XLSR-53 to predict IPA strings. Facebook-xlsr-espeak was trained on at least 100 hours from 26 languages selected from CommonVoice, while MultiIPA used 1,000 manually checked samples from 7 languages with consistent orthography-to-pronunciation mapping.

To test the out-of-domain performance of each model, we leverage test splits of datasets and languages that were not in any training set: DoReCo, the Kazakh Speech Dataset, and CommonVoice Tatar. Facebook-xlsr-espeak achieves the best results across all three datasets, likely due to its broader training coverage of 26 languages including non-European ones. Our eu21-50h model achieves competitive results on Tatar (57.3% PER; 17.2% PFER), approaching Facebook-xlsr-espeak (51.3% PER; 17.4% PFER), and outperforms MultiIPA on Tatar in PFER. As expected, all models degrade on DoReCo, which contains small and endangered languages far outside any training distribution.

Table 4
Out-of-domain PER and PFER (↓%) comparison

	PER			PFER		
Model	DoReCo	Kazakh	Tatar (CV)	DoReCo	Kazakh	Tatar (CV)
fb-xlsr-espeak	**51.7**	**51.7**	**51.3**	**22.5**	**19.3**	**17.4**
MultiIPA	59.6	71.6	62.9	<u>25.9</u>	20.7	17.4
eu21-fl (ours)	67.5	<u>61.6</u>	61.3	27.4	<u>20.4</u>	19.6
eu21-50h (ours)	<u>58.7</u>	62.2	<u>57.3</u>	26.0	21.1	<u>17.2</u>

5.3. Two-Step ASR Evaluation

We report results of the speech-to-ipa-to-grapheme pipeline. This second phonetic model is trained on five languages (English, Italian, German, Spanish, and French) and evaluated using the corresponding test set of FLEURS: Table 5 shows WER for each of the language-specific FSTs and for the combined single unified graph.

The system achieves the best results on English (17.7%) and German (19.7%), while French shows the highest WER (34.7%). The combined multilingual FST, which handles all five languages in a single decoding pass without any explicit language identification, achieves an overall WER of 30.3%.

Although the performance across languages remains below that of recent large-scale monolingual models, the results support the validity of the proposed approach. In particular, for constrained linguistic domains where a customized language model can be defined, combining a language-independent acoustic model with a domain-specific lexicon yields an effective and resource-efficient architecture. Such a design enables compact speech recognition systems with a reduced memory footprint, making them particularly suitable for deployment in resource-constrained environments.

Table 5
WER (↓%) of the WFST decoder on FLEURS test sets

	en	de	es	it	fr	all (combined)
WFST	17.7	19.7	25.6	25.6	34.7	30.3

6. Discussion

In the supervised setting, the results of the phonetic model show that increasing the training data yields modest gains (24.5% → 21.9% PER; 8.2% → 7.5% PFER), while in the zero-shot setting, eu21-fl outperforms eu21-50h in PER (43.3% vs. 44.9%). Indeed, while increasing the amount of training data improves supervised performance, the gains do not consistently transfer to unseen languages. In fact, the eu21-fl model achieves better zero-shot PER than eu21-50h, despite being trained on substantially fewer data. This suggests that exposure to larger amounts of speech from the same language set may encourage language-specific specialization, whereas a more balanced and homogeneous training corpus can preserve language-independent phonetic representations. This finding is consistent with the hypothesis that universal phonetic modeling depends not only on data quantity but also on the diversity and consistency of phonetic supervision.

In general, these results are consistent with the observation that transfer learning for speech-to-IPA reaches a performance plateau relatively early [13], and that quality can compensate for a substantial gap in data volume. Our models, trained on a curated subset of IPAPACK++, approach fb-xlsr-espeak, which was trained on at least 100 hours from 26 languages, on Tatar (57.3% vs. 51.3% PER; 17.2% vs. 17.4% PFER) and Kazakh (20.4% vs. 19.3% PFER). This is also proven by MultiIPA, which achieves competitive DoReCo PFER (25.9%) using only 1,000 manually verified samples from 7 languages. Fb-xlsr-espeak still remains the strongest out-of-domain baseline, benefiting from broader typological coverage.

Although zero-shot PER values remain relatively high for several languages, the corresponding PFER scores are substantially lower. This indicates that many recognition errors involve phonetically similar sounds rather than completely incorrect predictions: such behaviour is encouraging because it suggests that the model can capture broad phonological pattern. The relatively small gap in PFER between supervised and zero-shot conditions further supports the robustness of the phonetic space.

The WFST results are promising for low-resource languages, which often lack transcribed audio but may possess sufficient written text to train a language model. The proposed architecture fully decouples acoustic and linguistic modelling, so a single universal phonetic model could serve as the front-end for any language for which a text-based back-end can be constructed. English (17.7%) and German (19.7%) achieve the best results, while French (34.7%) performs the worst. The combined multilingual graph achieves 30.3% WER without any explicit language identification, demonstrating the viability of unified multilingual decoding. These results are noteworthy because the decoder operates without any explicit language identification component. The unified graph therefore performs two tasks simultaneously: selecting the most likely language model path and recovering the orthographic transcription; although absolute performance is still low, the phonetic representation generated by the acoustic model seems to be sufficiently discriminative to guide language selection implicitly. This result supports the feasibility of future multilingual and code-switching systems in which language boundaries are not known in advance.

6.1. Limitations and Future Work

The first limitation concerns the quality of training label. Both the eSpeak-phonemized FLEURS data and the IPAPACK++ dataset rely on automatic G2P conversion, rather than human-annotated IPA transcriptions. Despite the curation steps applied to IPAPACK++, G2P errors can propagate along the pipeline. The second limitation is the restricted language coverage. Training on 21 European languages is sufficient to demonstrate the approach, but does not reflect the typological diversity of the world's languages. As the comparison with fb-xlsr-espeak shows, broader coverage during training translates

directly into better out-of-domain performance on low-resource and endangered languages. Expanding the training set to include more language families is therefore a natural next step.

The two-step speech-to-IPA-to-grapheme pipeline also needs to be extended to more languages. The architecture is well suited to this: since the phonetic model is language-agnostic, adding a new language only requires text-based resources (a written corpus and a G2P tool to derive phoneme-level targets) without any additional transcribed speech. This makes the approach attractive precisely in low-resource scenarios, where written corpora may be available even when transcribed audio is not. A further promising direction is code-switching: a unified multilingual decoding graph, similar to the simple version demonstrated here, could handle language alternation without requiring explicit language identification.

7. Conclusion

In this paper, we have investigated IPA-based multilingual ASR using a fine-tuned XLSR-53 phonetic model and a hybrid speech-to-IPA-to-grapheme framework based on Weighted Finite State Transducer decoding. Our results show that training on a carefully curated set (IPAPACK++) achieves performance comparable to models trained on substantially larger datasets, underscoring the importance of labeling quality over dataset scale. The phonetic model demonstrates robust cross-lingual transfer, generalizing effectively to previously unseen languages in zero-shot settings. Moreover, a WFST decoder shows that accurate orthographic transcription can be generated using only text-derived resources, without requiring large quantity of transcribed speech. These findings highlight the potential of IPA-based approaches as a scalable and data-efficient pathway toward extending ASR systems to low-resource and previously unsupported languages.

Beyond ASR, the language-independent phonetic representation offered by IPA may enable applications in computer-assisted language learning, where accurate modeling of pronunciation and phonetic deviations is essential. By providing a common phonetic space across languages, the proposed framework could support pronunciation assessment. In addition, explicit phonetic representation may prove valuable for the analysis of pathological speech, facilitating the detection and characterization of atypical articulation patterns, speech impairments, and pronunciation errors.

Acknowledgments

This project is co-funded by the Digital Europe Programme under LLMs4EU, Grant Agreement No 101198470. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the granting authority can be held responsible for them.

Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic) to check grammar and spelling and rephrase selected passages. After using this tool, the authors reviewed and edited the content as needed and assume full responsibility for the content.

References

[1] D. Amodei, S. Ananthanarayanan, R. Anubhai, J. Bai, E. Battenberg, C. Case, J. Casper, B. Catanzaro, Q. Cheng, G. Chen, et al., Deep speech 2: End-to-end speech recognition in English and Mandarin, in: International conference on machine learning, PMLR, 2016, pp. 173–182.

[2] A. Babu, C. Wang, A. Tjandra, K. Lakhotia, Q. Xu, N. Goyal, K. Singh, P. Von Platen, Y. Saraf, J. Pino, et al., XLS-R: Self-supervised cross-lingual speech representation learning at scale, arXiv preprint arXiv:2111.09296 (2021).

[3] S. Feng, M. Tu, R. Xia, C. Huang, Y. Wang, Language-universal phonetic representation in multilingual speech pretraining for low-resource speech recognition, arXiv preprint arXiv:2305.11569 (2023).

[4] S. Yusuyin, T. Ma, H. Huang, W. Zhao, Z. Ou, Whistle: Data-efficient multilingual and crosslingual speech recognition via weakly phonetic supervision, IEEE Transactions on Audio, Speech and Language Processing (2025).

[5] Y.-A. Chung, J. Glass, Speech2vec: A sequence-to-sequence framework for learning word embeddings from speech, arXiv preprint arXiv:1803.08976 (2018).

[6] A. Baevski, Y. Zhou, A. Mohamed, M. Auli, wav2vec 2.0: A framework for self-supervised learning of speech representations, Advances in neural information processing systems 33 (2020) 12449–12460.

[7] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, A. Mohamed, Hubert: Self-supervised speech representation learning by masked prediction of hidden units, IEEE/ACM transactions on audio, speech, and language processing 29 (2021) 3451–3460.

[8] G. Synnaeve, Q. Xu, J. Kahn, T. Likhomanenko, E. Grave, V. Pratap, A. Sriram, V. Liptchinsky, R. Collobert, End-to-end asr: from supervised to semi-supervised learning with modern architectures, arXiv preprint arXiv:1911.08460 (2019).

[9] Q. Xu, A. Baevski, T. Likhomanenko, P. Tomasello, A. Conneau, R. Collobert, G. Synnaeve, M. Auli, Self-training and pre-training are complementary for speech recognition, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 3030–3034.

[10] S. Dalmia, R. Sanabria, F. Metze, A. W. Black, Sequence-based multi-lingual low resource speech recognition, in: 2018 IEEE international conference on acoustics, speech and signal processing (ICASSP), IEEE, 2018, pp. 4909–4913.

[11] V. Pratap, A. Sriram, P. Tomasello, A. Hannun, V. Liptchinsky, G. Synnaeve, R. Collobert, Massively multilingual ASR: 50 languages, 1 model, 1 billion parameters, arXiv preprint arXiv:2007.03001 (2020).

[12] Q. Xu, A. Baevski, M. Auli, Simple and effective zero-shot cross-lingual phoneme recognition, arXiv preprint arXiv:2109.11680 (2021).

[13] C. Taguchi, Y. Sakai, P. Haghani, D. Chiang, Universal automatic phonetic transcription into the international phonetic alphabet, arXiv preprint arXiv:2308.03917 (2023).

[14] P. Żelasko, L. Moro-Velázquez, M. Hasegawa-Johnson, O. Scharenborg, N. Dehak, That sounds familiar: an analysis of phonetic representations transfer across languages, arXiv preprint arXiv:2005.08118 (2020).

[15] J. Zhu, F. Samir, E. Chodroff, D. R. Mortensen, ZIPA: A family of efficient models for multilingual phone recognition, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2025, pp. 19568–19585.

[16] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, A. Bapna, Fleurs: Few-shot learning evaluation of universal representations of speech, in: 2022 IEEE Spoken Language Technology Workshop (SLT), IEEE, 2023, pp. 798–805.

[17] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, G. Weber, Common voice: A massively-multilingual speech corpus, in: Proceedings of the twelfth language resources and evaluation conference, 2020, pp. 4218–4222.

[18] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, I. Sutskever, Robust Speech Recognition via Large-Scale Weak Supervision, arXiv preprint arXiv:2212.04356 (2022).

[19] C.-J. Li, K. Chang, S. Bharadwaj, E. Yeo, K. Choi, J. Zhu, D. Mortensen, S. Watanabe, Powsm: A phonetic open whisper-style speech foundation model, arXiv preprint arXiv:2510.24992 (2025).

[20] S. Fong, M. Matassoni, A. Brutti, Speech LLMs in Low-Resource Scenarios: Data Volume Requirements and the Impact of Pretraining on High-Resource Languages, in: Interspeech 2025, 2025, pp.

2003–2007. doi:10.21437/Interspeech.2025-764.
[21] Z. Yao, L. Guo, X. Yang, W. Kang, F. Kuang, Y. Yang, Z. Jin, L. Lin, D. Povey, Zipformer: A faster and better encoder for automatic speech recognition, in: International Conference on Learning Representations, volume 2024, 2024, pp. 44440–44455.
[22] PHOIBLE 2.0, Jena, 2019. URL: https://phoible.org.
[23] P. Polák, S. Sagar, D. Macháček, O. Bojar, CUNI neural ASR with phoneme-level intermediate step for˜ non-native˜ SLT at IWSLT 2020, in: Proceedings of the 17th International Conference on Spoken Language Translation, 2020, pp. 191–199.
[24] H. Xue, Q. Shao, P. Chen, P. Guo, L. Xie, J. Liu, TranUSR: Phoneme-to-word transcoder based unified speech representation learning for cross-lingual speech recognition, arXiv preprint arXiv:2305.13629 (2023).
[25] B. Li, S.-y. Chang, T. N. Sainath, R. Pang, Y. He, T. Strohman, Y. Wu, Towards fast and accurate streaming end-to-end ASR, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 6069–6073.
[26] H. Gao, J. Ni, Y. Zhang, K. Qian, S. Chang, M. Hasegawa-Johnson, Zero-Shot Cross-Lingual Phonetic Recognition with External Language Embedding, in: Interspeech, 2021, pp. 1304–1308.
[27] V. Panayotov, G. Chen, D. Povey, S. Khudanpur, LibriSpeech: an ASR corpus based on public domain audio books, in: 2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2015, pp. 5206–5210.
[28] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, R. Collobert, MLS: A Large-Scale Multilingual Dataset for Speech Research, in: Proceedings of Interspeech, 2020, pp. 2757–2761. doi:10.21437/Interspeech.2020-2826.
[29] J. Zhu, C. Zhang, D. Jurgens, ByT5 model for massively multilingual grapheme-to-phoneme conversion, in: Proc. Interspeech 2022, 2022, pp. 446–450. URL: https://doi.org/10.21437/Interspeech.2022-538. doi:10.21437/Interspeech.2022-538.
[30] D. R. Mortensen, S. Dalmia, P. Littell, Epitran: Precision G2P for many languages, in: Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018), 2018.
[31] L. Paschen, F. Delafontaine, C. Draxler, S. Fuchs, M. Stave, F. Seifart, Building a time-aligned cross-linguistic reference corpus from language documentation data (DoReCo), in: Proceedings of the Twelfth Language Resources and Evaluation Conference, 2020, pp. 2657–2666.
[32] N. Kadyrbek, M. Mansurova, A. Shomanov, G. Makharova, The development of a Kazakh speech recognition model using a convolutional neural network with fixed character level filters, Big Data and Cognitive Computing 7 (2023) 132.
[33] A. Laptev, S. Majumdar, B. Ginsburg, CTC Variations Through New WFST Topologies, in: Interspeech 2022, 2022, pp. 1041–1045. doi:10.21437/Interspeech.2022-10854.
[34] M. Bernard, H. Titeux, Phonemizer: Text to phones transcription for multiple languages in python, Journal of Open Source Software 6 (2021) 3958.
[35] M. J. Gales, K. M. Knill, A. Ragni, S. P. Rath, Speech recognition and keyword spotting for low-resource languages: Babel project research at cued, in: Fourth International workshop on spoken language technologies for under-resourced languages (SLTU-2014), International Speech Communication Association (ISCA), 2014, pp. 16–23.
[36] A. Graves, S. Fernández, F. Gomez, J. Schmidhuber, Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks, in: Proceedings of the 23rd international conference on Machine learning, 2006, pp. 369–376.
[37] D. R. Mortensen, P. Littell, A. Bharadwaj, K. Goyal, C. Dyer, L. Levin, Panphon: A resource for mapping IPA segments to articulatory feature vectors, in: Proceedings of COLING 2016, the 26th international conference on computational linguistics: Technical papers, 2016, pp. 3475–3484.